Determine the derivative:

$$f(x) = \ln\left(\frac{x^2\sqrt{x+1}}{x-1}\right)$$

$$\Rightarrow f(x) = \left[\ln(x^2) + \ln(\sqrt{x+1})\right] - \ln(x-1)$$

$$f(x) = \left[\ln x^2 + \frac{1}{2}\ln(x+1)\right] - \ln(x-1)$$

$$f'(x) = 2\frac{1}{x} + \frac{1}{2}\left(\frac{1}{x+1}\right) - \frac{1}{x-1}$$

$$f'(x) = \frac{2}{x} + \frac{1}{2x+2} - \frac{1}{x-1}$$

$$f'(x) = \frac{4(x+1)(x-1) + x(x-1) - 2x(x+1)}{2x(x+1)(x-1)}$$

$$f'(x) = \frac{(4x^2-4) + (x^2-x) - (2x^2+2x)}{2x(x+1)(x-1)}$$

$$\boxed{f'(x) = \frac{3x^2 - 3x - 4}{2x(x+1)(x-1)}}$$

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Day 5